

1. Administrative & Study Identification

Full Study Title: A Study of Nemtabrutinib Plus Venetoclax vs Venetoclax + Rituximab (VR) in Second-line (2L) + Relapsed/Refractory (R/R) Chronic Lymphocytic Leukemia/Small Lymphocytic Lymphoma (CLL/SLL) (MK-1026-010/BELLWAVE-010). **Official Title:** A Phase 3, Open-label, Randomized Study to Compare the Efficacy and Safety of Nemtabrutinib (MK-1026) Plus Venetoclax Versus Venetoclax Plus Rituximab in Participants With Relapsed/Refractory Chronic Lymphocytic Leukemia/Small Lymphocytic Lymphoma Following at Least 1 Prior Therapy (BELLWAVE-010) **Short Title/Acronym:** BELLWAVE-010 (MK-1026-010) **Protocol Number:** MK-1026-010 **NCT Number:** NCT05947851 **Posting ID:** 140689145418742651 **Sponsor/Company:** Merck **Investigational Product:** Nemtabrutinib plus Venetoclax **Phase of Development:** Phase 3 **Trial Status:** Recruiting **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the safety, tolerability, and confirm the dose of nemtabrutinib in combination with venetoclax (Part 1); and to evaluate if the combination of nemtabrutinib plus venetoclax is superior to venetoclax + rituximab (VR) with respect to progression-free survival (PFS) (Part 2). **Secondary Objectives:** To evaluate undetectable minimal residual disease (MRD) in bone marrow at month 14, objective response rate (ORR), duration of response (DOR), overall survival (OS), and safety.

2.2 Background & Rationale To address the unmet medical need for more effective treatments in patients with relapsed or refractory (R/R) CLL/SLL after ≥ 1 prior therapy. Venetoclax plus rituximab is a standard of care, but nemtabrutinib—a BTK inhibitor targeting both wild-type and C481-mutant forms of BTK—combined with venetoclax aims to provide deeper and more durable responses, potentially offering superior efficacy.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Venetoclax + Rituximab) **Blinding:** Open-label **Randomization:** Randomized (Parallel-group, in Part 2)

3.2 Planned Enrollment & Duration Target \$N\$: 735 participants
Number of Sites: Global multicenter **Estimated Study Start:** 08-Aug-2023 **Estimated Primary Completion:** 27-Jun-2033

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Confirmed diagnosis of chronic lymphocytic leukemia/small lymphocytic lymphoma (CLL/SLL) and active disease clearly documented to initiate therapy. 2. Relapsed or refractory to at least 1 prior available therapy, and have at least 1 marker of disease burden. 3. Deletion (Del) (17p) status, tumor protein 53 (TP53) mutation status, and immunoglobulin heavy chain gene (IGHV) mutation status results required before randomization for Part 2 participants only. 4. Eastern Cooperative Oncology Group (ECOG) performance status of 0 to 2 within 7 days before randomization, and a life expectancy of at least 3 months. 5. Has the ability to swallow and retain oral medication. 6. Participants with HIV, HBV (undetectable DNA viral load after ≥ 4 weeks of antiviral therapy), or HCV (undetectable RNA viral load) who meet all other eligibility criteria. 7. Agrees to prescribed contraception or abstinence requirements for nemtabrutinib (12 days) and venetoclax (30 days); female participants of childbearing potential must have a negative highly sensitive pregnancy test and not be breastfeeding.

4.2 Exclusion Criteria 1. Has an active hepatitis B virus (HBV) or hepatitis C virus (HCV) infection (unless adequately treated/controlled as per inclusion criteria), or active infection requiring systemic IV therapy. 2. HIV-infected participants with a history of Kaposi's sarcoma, Multicentric Castleman's Disease, and/or AIDS-defining opportunistic infection in the past 12 months. 3. Has gastrointestinal (GI) dysfunction that may affect drug absorption. 4. Has diagnosis of Richter Transformation or active central nervous system (CNS) involvement by CLL/SLL. 5. Has a known additional malignancy that is progressing or has required active treatment within the past 2 years. 6. Has received prior B-cell lymphoma 2 inhibitor(s) (BCL2i) within ≤ 12 months before randomization, prior systemic anticancer therapy within 5 half-lives or 4 weeks (for monoclonal antibodies), or radiotherapy requiring corticosteroids within 2 weeks. 7. Has a known allergy/sensitivity to nemtabrutinib or contraindication to venetoclax/rituximab, or a history of severe bleeding disorders (e.g., hemophilia) or clinically significant cardiovascular disease. 8. Currently being treated with p-glycoprotein (P-gp) substrates with a narrow therapeutic index, CYP3A strong/moderate inducers, or CYP3A strong inhibitors. 9. Has received a live or live attenuated vaccine within 30 days before the first dose. 10. Has a known psychiatric or substance use disorder that would interfere with study requirements, or has not adequately recovered from major surgery.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Nemtabrutinib administered orally once daily in combination with Venetoclax. Venetoclax involves a standard ramp-up over 4 to 5 weeks to a target dose of 400 mg once daily. **Route of Administration:** Oral
Duration of Treatment: Continued until documented disease progression, unacceptable toxicity, or withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for oncology participants, including specific tumor lysis syndrome (TLS) prophylaxis during venetoclax initiation. **Prohibited:** Prior systemic anticancer therapy within 5 half-lives or 4 weeks, prior BCL2i within 12 months, live vaccines within 30 days, or concurrent use of strong CYP3A inhibitors/inducers.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Labs, Viral Load Checks, Mutation Status (Del 17p, TP53, IGHV required before Part 2 randomization)
Baseline	Day 0	Randomization, First Dose
Interim	Weeks 1-5 & subsequent	Safety Labs, Efficacy Assessment, Venetoclax Ramp-up (TLS monitoring), PK evaluations
End of Study		Disease Progression Final Vital Signs, Exit Interview, IP Accountability, Survival/MRD Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * **Part 1:** Number of Participants Experiencing Dose-Limiting Toxicities (DLTs) evaluated over 8 weeks after the first combination dose + 4 weeks follow up; Number of Participants Experiencing Adverse Events (AEs); and Number of Participants Discontinuing Study Treatment Due to AEs.
* **Part 2:** Progression-free survival (PFS) defined as the time from randomization to the first documented disease progression or death due to any cause. Assessed per 2018 International Workshop on Chronic Lymphocytic Leukemia (iwCLL) criteria by Blinded Independent Central Review (BICR).

7.2 Secondary & Exploratory Endpoints * **Part 2:** Undetectable minimal residual disease (MRD) rate in bone marrow as assessed by Central Laboratory. * **Part 2:** Overall Survival (OS). * **Part 2:** Objective Response Rate (ORR) per iwCLL Criteria 2018 as assessed by BICR.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection evaluating severity via toxicity scales, with special focus on DLTs. DLTs include Grade ≥ 3 nonhematologic toxicity, Grade 4 hematologic toxicity lasting >7 days, severe drug-induced liver injury, and missing $>25\%$ of doses due to drug-related AEs in the first 2 cycles. **Serious Adverse Events (SAEs):** Reported promptly based on protocol. Part 1 evaluated safety to confirm the Phase 2/3 dose. **Data Monitoring Committee (DMC):** External independent committee to oversee safety parameters and efficacy boundaries.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy outcome analysis (PFS). Safety Set for all AE/DLT evaluations. **Sample Size Justification:** 735 participants targeted to appropriately power statistical superiority claims between the nemtabrutinib + venetoclax arm and the VR arm. **Handling of Missing Data:** Handled per oncology SAP standards; time-to-event endpoints censored at the last evaluable assessment if progression/death is not observed.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Rigorous onsite and remote monitoring to verify BICR imaging and central lab MRD data. **Audit Trail:** Controlled eCRF locking and tracking of treatment protocol deviations.

11. Study Identifiers & Governance

Primary ID: MK-1026-010 **Registry Number:** NCT05947851 **Posting ID:** 140689145418742651 **Sponsor/Lead Organization:** Merck **Study Title:** BELLWAVE-010 **Official Regulatory Title:** A Phase 3, Open-label, Randomized Study to Compare the Efficacy and Safety of Nemtabrutinib (MK-1026) Plus Venetoclax Versus Venetoclax Plus Rituximab in Participants With Relapsed/Refractory Chronic Lymphocytic Leukemia/Small Lymphocytic Lymphoma Following at Least 1 Prior Therapy (BELLWAVE-010)

12. Clinical Framework (CLL/SLL Specific)

Primary Condition: Leukemia, Lymphocytic, Chronic, B-Cell; Leukemia, Chronic Lymphocytic, Small-Cell Lymphoma; Lymphoma, Small Lymphocytic; CLL/SLL **Diagnosis Threshold:** Active disease

meeting criteria to initiate therapy with at least 1 prior therapy line, Relapsed/Refractory **Medical Methodology:** Targeted dual inhibition (BTK inhibitor + BCL2 inhibitor) **Optimization Target:** Improved Progression-Free Survival (PFS) and achieving undetectable MRD

13. Study Procedures & Methodology

Management Pathway: Stepwise oral targeted therapy (nemtabrutinib starting cycle 1, venetoclax added cycle 2) **Education Protocol (HCP):** Training on protocol-specified venetoclax ramp-up to mitigate Tumor Lysis Syndrome (TLS) **Education Protocol (Patient):** Education on oral administration adherence, symptom reporting for signs of TLS, bleeding, or infection **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for staging and progression tracking

14. Observation & Follow-Up Schedule

Total Duration: Up to 10 years (Completion approx. 2033) **Onsite Visit Cadence:** Increased monitoring frequency during the initial 28-day cycle and 4-week venetoclax ramp-up, followed by routine oncology follow-up **Remote Monitoring Frequency:** Intermittent phone checks per clinical trial schedule **Checklist Review Interval:** Routine continuous assessment of AEs and concomitant medications at every protocol-defined visit

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				